English translation of the original German document

Examination regulations for the Master's degree programme Sustainability Transformation in Engineering and Management of the Faculty 14 Industrial Engineering and Management (WI) of the Technische Hochschule Mittelhessen (THM) dated 14 January 2025 - Version 1

Authorisation:

In accordance with Section 43 (5) of the Hessian Higher Education Act (HessHG) in the version dated 14 December 2021 (GVBl. p. 931), last amended on 10 October 2024 (GVBl. 2024, No. 56), I hereby approve the following examination regulations for the Master's degree programme Sustainability Transformation in Engineering and Management.

For the Executive Committee

Giessen, 6 February 2025

Prof. Dr. Matthias Willems,
Pesident of the Technischen Hochschule Mittelhessen

Preface:

In accordance with Section 50 (1) No. 1 of the HessHG in the version dated 14 December 2021 (GVBl. 2021, 931), last amended on 10 October 2024 (GVBl. 2024, No. 56), the Faculty Council of the WI Faculty of THM adopted the examination regulations for the Master's degree programme Sustainability Transformation in Engineering and Management on 14 January 2025. Part I contains the General Provisions for Master's Examination Regulations of THM dated 14 January 2015 (AMB 01/2015), last amended on 21 February 2024 (AMB 16/2024), and is supplemented by the subject-specific provisions in Part II.

Resolution FBR	Resolution Senate	Approval President	Effective Date / Publication
14 Janury 2025	05 February 2025	06 February 2025	01 June 2025 / AMB 42/2025

Part I
General Provisions

The General Provisions for Master's Examination Regulations of the THM of 14 January 2015 (AMB 01/2015), last amended on 21 February 2024 (AMB 16/2024), published in the Official Bulletin of the THM, apply.

Part II

Subject-specific Provisions

Table of Contents

§ 1 Scope, General Information, Study Objectives

(1) The subject-specific provisions regulate the content and requirements of the Master's degree programme Sustainability Transformation in Engineering and Management with the degree 'Master of Science' of the Department of Industrial Engineering and Management (WI).

(2) The aim of the Master's degree programme is to enable students to become change agents who are able to identify fields of action relevant to sustainability in companies from different sectors, institutes, NGOs, etc. and to develop and implement technical or organisational solutions. Based on their interdisciplinary qualifications, they develop sustainability strategies and solutions, which they implement, communicate and optimise. In doing so, they facilitate constructive dialogues between decision-makers and stakeholders. Graduates are able to work successfully in interdisciplinary teams and establish sustainability strategies at all levels of their company or organisation.

(3) The degree programme is highly application-oriented.

§ 2 Teaching and Learning Concept: Challenge Based Learning

(1) The Master's degree programme Sustainability Transformation in Engineering and Management uses the teaching and learning concept of Challenge Based Learning (CBL). In semesters 1-3, students complete an interdisciplinary or transdisciplinary project based on this approach totalling 10 CP, which is geared towards a specific key topic in the area of sustainability (see Appendix 2)
Semester 1: Energy and resource supply and provision
Semester 2: Sustainable cities and communities | industry, innovation, infrastructure
Semester 3: Sustainable consumption | Sustainable production

(2) In these project modules, students develop questions independently in project teams, taking into account their respective subject requirements and the interdisciplinary composition of the team, and derive motivating challenges from the main topics which they work on over the course of the semester in accordance with the teaching and learning concept.

(3) A project team consists of at least two students and is put together by the module coordinator according to the project requirements and the individual competences of the students as well as on the basis of their initial qualifications in an interdisciplinary manner. The composition of the group will be communicated to the students in good time and in an appropriate manner at the beginning of the course. Changing the project team and changing the individual assignment is only possible in exceptional cases and with the approval of the module coordinator.

(4) The project-related supervision of students in a Challenge-Based Learning module (CBL) by lecturers (with tutorial support if necessary) from the participating departments covers both subject-related and interdisciplinary, process-related aspects:

1. Professional support for self-study includes the provision of subject-specific resources such as learning videos, literature and data as required, as well as accompanying meetings to work on specific issues relevant to the project. If required, specialist input and methodological support, possibly also from other experts, is offered in the form of mini-lectures in order to increase feasibility, scientific foundation and innovation potential. The duration of the mini-lectures (usually 45 or 90 minutes) will be announced to students in good time and in an appropriate manner.

2. As part of the process-accompanying coaching, teachers support teamwork by asking specific questions that encourage critical thinking and promote self-organised project manage-

ment. Through regular reflection and feedback sessions, teachers promote the ability to reflect on oneself and the team and to give and accept constructive feedback. The projects should clearly demonstrate how a challenge is dealt with. In addition, where appropriate and possible, other cooperation partners from society and companies will be actively involved in working on the challenges as practitioners, clients, consultants or in similar roles.

§ 3 University Entrance Qualifications

(1) The Master's degree programme in Sustainability Transformation in Engineering and Management builds consecutively on a completed first professionally qualifying university degree in an engineering, technical, business or social science subject with a standard period of study of 6 semesters (180 CP). The Admission Board shall decide whether this requirement has been met.

(2) Admission to the Master's degree programme in Sustainability Transformation in Engineering and Management is only possible if proof of a first professionally qualifying university degree in accordance with paragraph 1 can be provided.

§ 4 Admission requirements

(1) Admission to the Master's degree programme Sustainability Transformation in Engineering and Management requires the fully completed ‘Application for admission to the Master's degree programme Sustainability Transformation in Engineering and Management’ via the portal specified by THM within the application deadline specified by THM, including the following documents:

1. Proof of the first professionally qualifying university degree according to § 3 para. 1 by means of
   a) Degree certificate,
   b) Diploma Supplement and
   c) Transcript of Records or comparable proof.
   If these final documents of the first professionally qualifying university degree cannot be submitted by the end of the application deadline, § 5 para. 4 applies.

2. Foreign applicants who have obtained their first professionally qualifying university degree in a non-German-language degree programme: Proof of sufficient knowledge of German at level A1 of the Common European Framework of Reference for Languages.

3. Proof of sufficient knowledge of English at level B2 of the Common European Framework of Reference for Languages, usually proven by the Cambridge First Certificate, the TOEFL test, TOEIC test or IELTS test. The examination board decides on the recognition of proof of sufficient language skills.

(2) All documents to be submitted in accordance with para. 1 must be issued in German or English. If documents are issued in a language other than German or English, an officially recognised translation of the documents into German or English must be obtained and submitted with the original documents. If documents are issued in English, these documents will be reviewed by the Admissions Committee.

§ 5 Admission decision

(1) The university shall decide on the recognition of foreign or equivalent degrees and university entrance qualifications in accordance with the statutory provisions and the requirements of the Conference of University Rectors and the Conference of Ministers of Education and Cultural Affairs.

(2) Admission to the Master's degree programme in Sustainability Transformation in Engineering

and Management is granted to those who fulfil the admission requirements according to § 3 and the admission requirements according to § 4. The Admission Board decides on the fulfilment of all subject-specific admission requirements. It may consult other professors or academic staff for a preliminary examination of the application documents. In cases of doubt, it can request further documents from the applicant for the decision, e.g. certificates of previous relevant professional activities as well as further education and training courses.

(3) Admission is granted by the President (Registrar's Office) of the Technische Hochschule Mittelhessen. The applicant must be notified in writing of any rejection decisions. The decision must include information on legal remedies.

(4) Applicants who have not yet completed their first professionally qualifying university degree programme by the application deadline must submit a detailed certificate on the status of this degree programme instead of the degree documents required in Section 4 (1) No. 1 in order to be granted conditional admission to the Master's degree programme.
This certificate must show the following:

- all achievements to date in the first professionally qualifying degree programme,
- the number of credit points currently achieved and the total number of credit points to be achieved; at least 80% of the total number of credit points to be achieved must already have been earned.

Proof of completion of the first professionally qualifying university degree must be provided by the end of the enrolment period set by THM for the Master's degree programme in Sustainability Transformation in Engineering and Management. Failure to do so will result in conditional admission being cancelled.

§ 6 Matriculation

Admitted applicants must enroll by the deadline set by THM. Applicants can only be enrolled if they fulfil all enrolment requirements in accordance with Section 3 of the THM Enrollment Statutes as amended.

§ 7 Master's Degrees and Certificates

Upon successful completion of the Master's degree programme in 'Sustainability Transformation in Engineering and Management', the Master's degree 'Master of Science (M.Sc.)' is awarded with a certificate in accordance with Appendix 4.

§ 8 Standard period of study, duration and structure of the programme, modules, Master's thesis, language, last repetition of written examinations

(1) The standard period of study for the Master's degree programme is four semesters; this corresponds to two academic years. To successfully complete the Master's examination, the modules listed in the module overview in Appendix 1 must be successfully completed. The entire programme comprises 120 credit points.

(2) The Master's degree programme is offered in the winter and summer semesters.

(3) The language of instruction and examination is English. Other languages and the type of examinations are specified in the module handbook (Appendix 2)

(4) The modules to be completed are generally to be taken from the Master's degree programme as listed in Appendix 1 or 2. Alternatively, identical or equivalent modules can also be taken from the modules offered in other degree programmes at THM. Any failed attempts will be recognised. §§ Sections 11 to 14 of the General Provisions (Part I of the Examination Regulations) apply.

(5) Students can be admitted to the Master's thesis if they have successfully completed modules totalling at least 80 CP from the modules of the first three semesters as well as the modules 'Transformation Fields - Interdisciplinary Cooperation Project 1', 'Transformation Fields - Interdisciplinary Cooperation Project 2' and 'Transformation Fields - Interdisciplinary Cooperation Project 3'. In justified exceptional cases, the Master's thesis may be started before the required 80 CrP have been completed. The Admission Board will decide on this. The Master's thesis including colloquium has a total scope of 30 credit points. The Master's thesis takes 26 weeks to complete.

(6) Students must present and defend their work in a colloquium for the Master's thesis. In order to be admitted to the colloquium, all modules of the curriculum in Appendix 1 must be successfully completed with the exception of the 'Master's thesis' module. The written part of the 'Master's thesis' module must be passed.

(7) For elective modules, binding registration is required before the end of the lecture weeks of the previous semester. If there are fewer than 6 binding registrations, there is no entitlement to attend the course.

(8) In an 'accelerated procedure', elective modules that have not yet been offered, which address current topics and are of interest to students, can be offered without a change to the examination regulations being made in advance. The procedural requirements for this are regulated in Appendix 2.

(9) In the case of examinations and partial examinations for which proof of performance is provided in the form of a written examination, the last repetition may be carried out as an oral examination in accordance with Section 7 of the General Provisions upon written application by the candidate. This does not apply to the Master's thesis or the examination performance of the modules 'Transformation Fields - Interdisciplinary Cooperation Project 1', 'Transformation Fields - Interdisciplinary Cooperation Project 2' and 'Transformation Fields - Interdisciplinary Cooperation Project 3'. The oral examination for a last repetition can only be taken once during the degree programme. The admission board decides on the application.

§ 9 Obligation to pay costs

No fees are charged for the Master's degree programme in accordance with Section 20 (5) HessHG. The obligation to pay the semester fee in accordance with Section 83 (3) HessHG, the administrative fee in accordance with Section 62 HessHG and fees and contributions in accordance with other statutory provisions remains unaffected.